

CS 2614: Computer Organization

Lab 5

Spring 2025

2-Bit Multiplier

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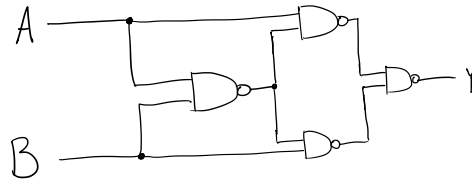
The UNIVERSITY of OKLAHOMA
Gallopy College of Engineering
School of Computer Science

Pre-Lab Exercise

1. Draw the logic diagram for the XOR function (using NAND gates only).

$$Y = A'B + AB'$$

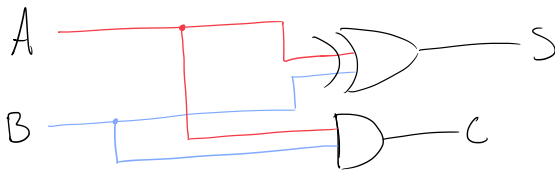
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0



2. Compute the following 2-bit binary multiplication (show all steps):

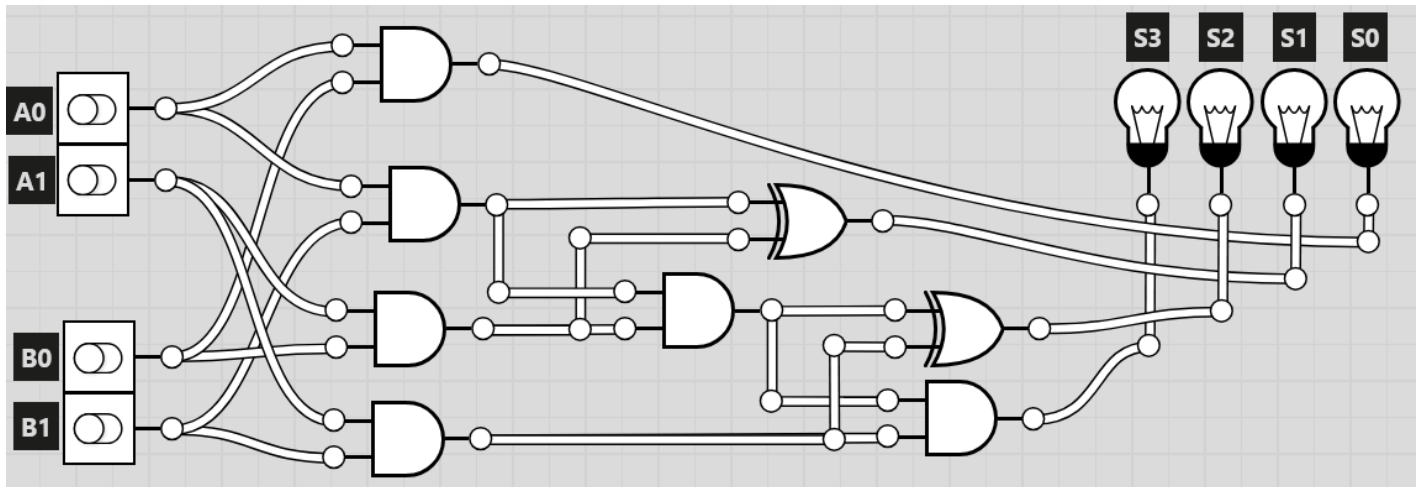
$$\begin{array}{r} 11 \\ * 10 \\ \hline 00 \\ 110 \\ \hline 110 \end{array} \Rightarrow \begin{array}{r} 3 \\ \times 2 \\ \hline 6_{10} \\ \downarrow \\ 110_2 \end{array}$$

3. Draw the logic diagram and truth table for a half-adder.



A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

4. Design a 2-bit binary multiplier [Hint: design in Q3 can be useful].



False = 0 True = 1

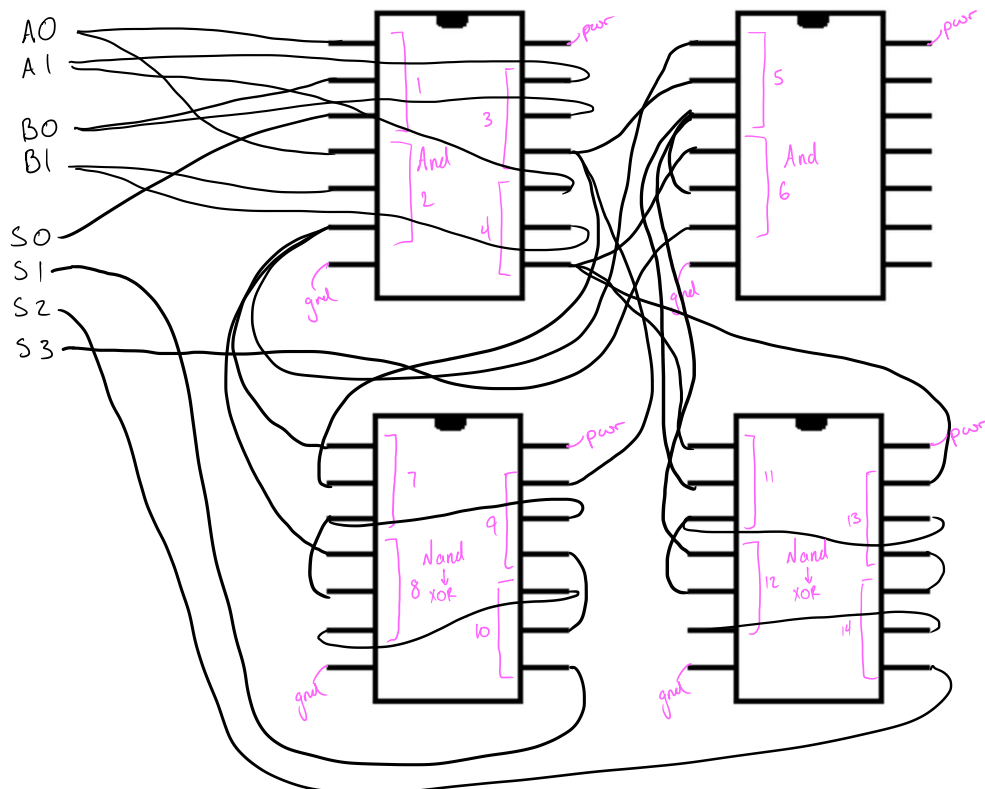
The organization is really bad, but it's right so oh well

A and B are 2 bit numbers				S is a 4 bit number.			
X0 is the least significant and X1 is the most significant				S0 is the least significant and S3 is the most significant			
A0	A1	B0	B1	S3	S2	S1	S0
FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE
FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE
FALSE	FALSE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE
FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
FALSE	TRUE	FALSE	TRUE	FALSE	TRUE	FALSE	FALSE
FALSE	TRUE	TRUE	FALSE	FALSE	FALSE	TRUE	FALSE
FALSE	TRUE	TRUE	TRUE	FALSE	TRUE	TRUE	FALSE
TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
TRUE	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE	FALSE
TRUE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	TRUE
TRUE	FALSE	TRUE	TRUE	FALSE	FALSE	TRUE	TRUE
TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
TRUE	TRUE	FALSE	TRUE	FALSE	TRUE	TRUE	FALSE
TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE
TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	TRUE

Remember to upload this to Canvas.

In-Lab Exercise

5. Draw a circuit from Problem 4 using the chips given below (label the chips and each input/output, some chips may not be needed)



6. Implement the circuit of Problem 5 on the breadboard.

